

# CASE OLLILA

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Quantitative finance researcher with a Finance degree and a portfolio of empirical market-research systems: options pricing in q/kdb+ and OCaml, volatility surfaces, hedging backtests, prediction-market calibration, Monte Carlo simulation, and systematic strategy infrastructure. Strong bias toward reproducibility, statistical validation, noisy-data hygiene, and measuring model failure rather than optimizing headline backtests.

## EDUCATION

**Colorado State University — B.S. Business Administration, Concentration in Finance** | August 2026

Coursework: Statistics with Business Applications; Analytics & AI in Business; Derivative Securities; Debt Securities Analysis; Advanced Portfolio Management; Financial Markets & Institutions; Principles of Investments

## QUANTITATIVE RESEARCH PROJECTS — INDEPENDENT WORK, MAY 2024 – AUGUST 2026

### Monte Carlo Option Pricer | q/kdb+

- Built a fully vectorized Monte Carlo engine in idiomatic q pricing European, Asian, and knock-out barrier options; paths stored as an  $n_{\text{steps}} \times n_{\text{paths}}$  matrix with time recursion as a single prefix-sum scan — no loop ever touches a path.
- Sustained ~956K paths/sec plain and ~1.43M paths/sec with antithetic variates (1.996× measured variance reduction), roughly 50–75M simulated price points per second end-to-end at 1M paths × 52 steps.
- Validated against closed-form Black-Scholes within 0.8 standard errors across five strikes; 30-check suite covering path generation, payoff ordering, and  $1/\sqrt{N}$  convergence (SE  $\propto \sqrt{N}$  flat across path counts).

### Implied Volatility Surface & SVI Calibration | Python, NumPy, SciPy

- Built an options-surface pipeline from 11,169 SPY contracts: cleaned noisy quotes, recovered forwards from put-call parity, inverted Black-Scholes volatility, and calibrated raw SVI across 12 expiries.
- Achieved 15.2 vol-bp median fit error with all 12 slices passing butterfly-arbitrage diagnostics and every adjacent maturity pair passing calendar-arbitrage checks; pricing, Greeks, IV solver, and diagnostics from scratch.
- Added multi-start nonlinear calibration, analytic Jacobian, vega weighting, reproducible offline snapshots, and 143 tests including end-to-end recovery of a known synthetic surface.

### Prediction Market Calibration Research — Polymarket & Kalshi | Python, SQL

- Built an end-to-end research pipeline spanning historical backfill, settlement reconciliation, calibration analysis, event-clustered inference, and cost-aware tradeability logic across 89,291 price snapshots covering 351 MLB games.
- Compared 21,733 paired hourly observations across 234 events; found a Polymarket-vs-Kalshi Brier-score difference of +0.00006 with 95% event-clustered CI  $[-0.00057, +0.00070]$ , rejecting a spurious favorite-longshot effect created by row-level dependence.
- Designed conservative event matching, raw-payload storage, bid/ask-aware execution assumptions, clustered bootstrap inference, and threshold sweeps to separate signal from fees, spreads, and repeated observations.

### Delta/Gamma Hedging Backtest | R

- Built a path-dependent hedging study over 823 overlapping three-month SPY windows (2010–2026): Black-Scholes pricing, Greeks, transaction costs, rebalancing, bootstrap inference, and Monte Carlo validation from scratch.
- Measured delta hedging reducing terminal P&L standard deviation from \$1,409 to \$373 per contract and delta-gamma to \$33; presented the variance-return tradeoff rather than hedging as a free improvement.
- Validated with 212 assertions, finite-difference Greeks, Black-Scholes PDE checks,  $1/\sqrt{m}$  convergence, and a real-data variance-gap regression ( $R^2 = 0.37$ ).

### ExaltedFable Agent Trading Lab | Python, SQLite, Alpaca Paper API

- Built a systematic strategy research lab with schema-validated trade proposals, deterministic risk approval, dry-run/paper execution, SQLite audit logs, SPY benchmarking, and multi-strategy tournament comparison.
- Separated signal generation from execution so strategy/LLM outputs cannot bypass position, cash, turnover, or asset constraints; linked proposals, risk decisions, orders, snapshots, and reports by experiment run.

## EXPERIENCE

**Private Wealth Intern — KW Capital Advisors / Ameriprise Financial, Boulder, CO** | Jun–Aug 2026

- Supported private wealth advisory workflows through client-meeting preparation, investment and planning materials, market research, and advisor follow-up organization.
- Observed portfolio reviews covering asset allocation, risk tolerance, liquidity, retirement income, tax-aware planning, and market conditions for affluent clients.

## TECHNICAL SKILLS

Python; R; q/kdb+; OCaml; C++; JavaScript/Node.js; SQL/SQLite; Git/GitHub; NumPy; SciPy; statistical inference; Monte Carlo; bootstrapping; event studies; options pricing; volatility surfaces; backtesting; risk controls; market-data pipelines

## COMPETITIVE PERFORMANCE & ENTREPRENEURSHIP

- Professional Fortnite competitor (2018–2021): competed in a high-feedback environment demanding rapid decisions, adaptation, deliberate preparation, and consistent performance under pressure.
- Built a 243K-subscriber YouTube channel solo (30.4M lifetime views, six videos past 1M), applying audience analytics, experimentation, and performance optimization to grow a large digital platform.